Stage 1: Intent Input

Natural-language intents (one sentence each)

Zone inventory (zone → router, prefixes)

Topology $G = (R, E)$ (Topology Zoo, ClassBench prefixes)

Output: $s_1 \ldots s_n$ and inventory

Stage 2: Per-Sentence LLM Translation

Qwen3-8B (Q4_K_M) llama.cpp · greedy JSON output

Validator type · zones services

One typed object τ_i per sentence (5 types)

invalid → re-ask (≤ 2)

1 call per sentence
7 placeholder examples

Output: $\tau_1 \ldots \tau_n$

Stage 3: Compiler and Rule Blocks

Precedence compiler

Zone policy Π (first match, default deny)

Prefix-pair blocks $B_{p,p'}$ · cost c_b

disjoint in header space · idempotent · 1-entry deny block for deny-only pairs

Output: blocks $\{B_b, c_b\}$ · paths $\mathscr{E}(s, d)$

Stage 4: Capacity-Aware ECMP Placement

All equal-cost paths $\mathscr{E}(s, d)$ (up to 28)

Router capacity C_r (ACL / TCAM entries)

Placement ILP
$x_{br} \in \{0, 1\}$
SciPy · HiGHS (≤ 30 s)

baselines: greedy · ingress egress

witness → re-place

Output: placement x^*

Stage 5: Render and Verify

Render iptables-restore

Independent parser (no access to blocks)

Network-wide Z3 check
every (s, d)
and every path π
full 5-tuple header

unsat → deploy

sat → packet witness

Output: verified router configurations

Evaluation: Emulated Testbed (GNS3 VM)

FRR 10.2.1 routers · OSPF · ECMP

iptables FORWARD per router

Zone hosts (one per zone)

Probes: UDP · TCP-SYN · ICMP

conformance · leaks · convergence · push time · update latency

Detail: Typed IR

allow(s, d, S)

allow_except(s, d, b, X)

block(s, d)

isolate(s, A)

public(d, S)

precedence: block ≻ except ≻ allow ≻ isolate ≻ public

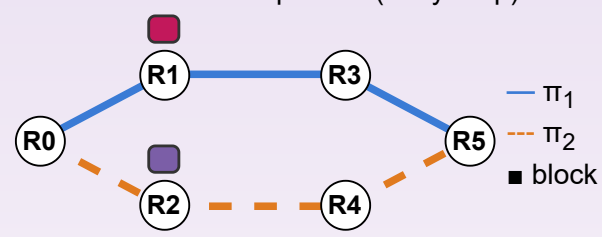
Detail: Placement ILP

$$\min \Sigma_{b,r}\, c_b\, x_{br}\, (1 + \alpha\, h_{br})$$

$$\Sigma_{r \in \pi}\, x_{br} \geq 1 \quad \forall\, b, \forall\, \pi \in \mathscr{E}(b)$$

$$\Sigma_b\, c_b\, x_{br} \leq C_r \quad \forall\, r$$

$\alpha = 0.05$ · h = hop index (early drop)



Detail: Z3 Verification

header h (bit-vectors)

src 32 bit	dst 32 bit	proto 8	port 16

$\text{dev}_r(h)$ = first match of the parsed rules of r, else forward

$\text{deliv}_\pi(h) = \wedge_{r \in \pi}\, \text{dev}_r(h)$

SAT?
$\text{src} \in P(s) \wedge \text{dst} \in P(d)$
$\wedge\, \text{deliv}_\pi(h) \neq \Pi(s, d, h)$

unsat $\forall\, (s, d), \pi$ ⇒ deployment ≡ Π

sat ⇒ packet witness + path π

one query per zone pair and per equal-cost path; prefix membership as masked equality